

DCPAM: A Dual-Camera-Based Point-to-Axis Measurement Model with Laser Reference Line

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Abstract—In large-scale rotating machinery such as turbines, generators, and compressors, shaft alignment accuracy is critical to operational balance and equipment lifespan. Traditional steel-wire alignment methods suffer from wire sag, vibration sensitivity, and limited measurement range. This paper proposes DCPAM (Dual-Camera-based Point-to-Axis Measurement), a non-contact measurement model that replaces the steel wire with a laser reference line and employs a dual-camera system to reconstruct the three-dimensional laser axis. The model computes the perpendicular distance from any target point to the reconstructed laser axis through a pipeline of pixel extraction, camera-to-world coordinate transformation, mirror-reflection virtual-image mapping, and cross-product distance calculation. Two parameter-tuning strategies are presented: DCPAM-CV, an experience-driven classical computer-vision workflow, and DCPAM-CNN, a convolutional-neural-network-based approach. A closed-form error propagation analysis and Monte Carlo validation quantify the measurement uncertainty under micrometer-level perturbations.

Index Terms—Shaft alignment, laser measurement, dual-camera system, point-to-axis distance, error propagation, computer vision.

I. Introduction

IN the power generation, marine, and aerospace industries, large rotating equipment—turbines, generators, compressors—features long shafts whose installation accuracy directly governs operational balance and service life. Shaft alignment (or “centreline finding”) is therefore a critical step during installation and commissioning of such machinery [1].

The prevailing industrial practice remains the steel-wire method, in which a taut piano wire is stretched along the nominal shaft axis and the radial distance from each bearing journal to the wire is measured with a micrometer or dial indicator. Despite its simplicity, the steel-wire method has well-known drawbacks: gravitational sag introduces systematic curvature that must be compensated; the wire is sensitive to air currents and mechanical vibration; and the achievable measurement range is limited by wire tension and sag correction accuracy.

Laser-based alignment offers a promising alternative. By replacing the steel wire with a laser beam as the reference axis, the problems of sag and vibration sensitivity are eliminated. However, the laser beam itself is invisible in free space; its position can only be observed where it strikes a receiving surface. This paper proposes DCPAM (Dual-Camera-based Point-to-Axis Measurement), a

model that employs two cameras—each observing the laser spot on its own receiving screen—to reconstruct the three-dimensional laser axis and compute the perpendicular distance from an arbitrary target point to that axis.

The contributions of this work are as follows: (1) A complete mathematical formulation of the dual-camera laser measurement pipeline, from pixel extraction to point-to-axis distance computation. (2) Two parameter-tuning strategies: DCPAM-CV (classical computer vision) and DCPAM-CNN (convolutional neural network). (3) A closed-form error propagation analysis with Monte Carlo validation. (4) A tilt-deviation rejection model that determines the maximum allowable probe inclination for a given target accuracy.

II. DCPAM Model Overview

The DCPAM model takes as input the images from two cameras and the length of a measurement tool bar, and outputs the perpendicular distance from the target point to the laser reference axis:

$$h_t = \text{DCPAM}(I_{C_1}, I_{C_2}, l_T) \quad (1)$$

where h_t is the target-point-to-axis distance, I_{C_1} and I_{C_2} are the images captured by cameras C_1 and C_2 respectively, and l_T is the length of the tool bar (side gauge).

Two parameter-tuning strategies are considered:

DCPAM-CV. An experience-driven workflow in which the operator iteratively adjusts CV algorithm parameters (thresholds, filter sizes, fitting methods), evaluates on a test set, and manually refines the pipeline until the target accuracy is achieved.

DCPAM-CNN. A deep-learning approach that trains a convolutional neural network to directly regress the laser spot centre from raw images, replacing hand-tuned CV processing.

III. DCPAM-CV: Classical Vision Pipeline

This section details the complete measurement pipeline of the classical computer-vision variant. Fig. ?? illustrates the data flow from raw images to the final distance measurement.

A. Coordinate System Definitions

We define three coordinate frames, all right-handed:

Camera coordinate frame $C1$. Origin at the optical centre of camera C_1 ; Z -axis aligned with the optical axis. A point P in this frame is denoted $P^{C1} = (x_1, y_1, z_1)^T$.

Camera coordinate frame $C2$. Origin at the optical centre of camera $C2$; Z -axis aligned with its optical axis. A point in this frame is denoted $P^{C2} = (x_2, y_2, z_2)^T$.

Device coordinate frame D . By convention, $D \equiv C1$ —the device frame coincides with the front camera coordinate frame. All final computations are performed in D .

The optical centres of the two cameras in their own frames are trivially $O_{C1} = (0, 0, 0)^T$ and $O_{C2} = (0, 0, 0)^T$. Note that this is a definitional identity: the intrinsic parameters (focal length, principal point, distortion) govern the projection relationship but do not affect the location of the optical centre in the camera's own coordinate frame.

B. Key Points

We define the following key points (all expressed in $C1$ unless otherwise noted):

P_{real}^f : the laser spot on the front receiving screen (real image in $C1$).

P_{real}^f : the virtual-image point obtained by rotating P_{real}^f about the mirror hinge (virtual image in $C1$).

P_{real}^{b-C2} : the laser spot on the rear receiving screen (real image in $C2$).

$P_{\text{virtual}}^{b-C2}$: the virtual-image point of the rear screen (virtual image in $C2$).

P_{virtual}^b : $P_{\text{virtual}}^{b-C2}$ transformed into $C1$.

P_T : the target point whose distance to the laser axis is to be measured.

C. Laser Spot Centre Extraction

Let the pixel coordinates of the laser spot in $C1$ and $C2$ be (u_1, v_1) and (u_2, v_2) , with homogeneous representations

$$\mathbf{p}_1 = \begin{bmatrix} u_1 \\ v_1 \\ 1 \end{bmatrix}, \quad \mathbf{p}_2 = \begin{bmatrix} u_2 \\ v_2 \\ 1 \end{bmatrix}. \quad (2)$$

D. Pixel-to-Camera Coordinate Transformation

Let K_1 and K_2 be the intrinsic matrices of cameras $C1$ and $C2$, and let Z_s be the known distance from the camera optical centre to the receiving screen along the optical axis. The back-projection from pixel coordinates to camera coordinates is

$$P_{\text{real}}^f = Z_s K_1^{-1} \mathbf{p}_1, \quad P_{\text{real}}^{b-C2} = Z_s K_2^{-1} \mathbf{p}_2 \quad (3)$$

Dependencies. Camera calibration provides K_1, K_2 ; the tool geometry provides Z_s .

Error sources. The accuracy of Z_s depends on the mechanical precision of the probe fixture.

E. Real-to-Virtual Point Transformation

The receiving screen is mounted on a mirror that can rotate about a hinge axis. To reconstruct the “unfolded” laser path, each real image point must be rotated to its virtual position.

In frame $C1$, let the rotation centre be $R_{O1} = (x_{O1}, 0, z_{O1})^T$, and let θ be the rotation angle (counter-clockwise from Z towards X in a right-handed frame). Then

$$P_{\text{virtual}}^f = H_1 \cdot P_{\text{real}}^f \quad (4)$$

where H_1 is a 4×4 rigid-body transformation matrix combining rotation and translation:

$$H_1 = \begin{bmatrix} \cos \theta & 0 & \sin \theta & x_{O1}(1 - \cos \theta) - z_{O1} \sin \theta \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & z_{O1}(1 - \cos \theta) + x_{O1} \sin \theta \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

The transformation represents a rotation about an axis parallel to Y that passes through R_{O1} .

Similarly, the rear real-image point is rotated in $C2$ about the centre $R_{O2} = (x_{O2}, 0, z_{O2})^T$ by angle 2θ :

$$P_{\text{virtual}}^{b-C2} = H_2 \cdot P_{\text{real}}^{b-C2} \quad (6)$$

Dependencies. Geometric dimensions R_{O1}, R_{O2} , and the rotation angle θ .

Ideal-case simplification. Under ideal assembly, $z_{O1} = z_{O2} = Z_s$.

F. $C2$ -to- $C1$ Coordinate Transformation

The virtual-image point $P_{\text{virtual}}^{b-C2}$ is expressed in frame $C2$ and must be converted to $C1$:

$$P_{\text{virtual}}^b = T_{C2 \rightarrow C1} \cdot P_{\text{virtual}}^{b-C2} \quad (7)$$

The transformation matrix is computed from the extrinsic parameters of the two cameras with respect to a common world frame W :

$$T_{C2 \rightarrow C1} = E_1 \cdot E_2^{-1} \quad (8)$$

where E_1, E_2 are the 4×4 extrinsic matrices (world-to-camera) of $C1$ and $C2$ respectively.

Equivalently, if the relative pose is parameterised as rotation R_{21} and translation t_{21} (from $C1$ to $C2$), then

$$R_{12} = R_{21}^T \quad (9)$$

$$t_{12} = -R_{21}^T t_{21} \quad (10)$$

and $P^{C1} = R_{12} P^{C2} + t_{12}$.

Dependencies. Stereo camera calibration (extrinsic parameters).

G. Point-to-Axis Distance Computation

With $P_{\text{virtual}}^f, P_{\text{virtual}}^b$, and P_T all expressed in $C1$, the perpendicular distance from P_T to the laser axis (the line through P_{virtual}^f and P_{virtual}^b) is given by the classical cross-product formula:

$$H = \frac{\|\vec{L} \times \vec{w}\|}{\|\vec{L}\|} \quad (11)$$

where

$$\vec{L} = P_{\text{virtual}}^f - P_{\text{virtual}}^b \quad (12)$$

$$\vec{w} = P_T - P_{\text{virtual}}^b \quad (13)$$

The vector $\vec{L}$ is the direction vector of the reconstructed laser axis in frame $C1$.

Dependencies. The target point P_T is determined by the tool-bar geometry and the measurement configuration.

IV. Precision Analysis

A. End-to-End Matrix Formulation

Combining the transformations from Sections III-D–III-F, the two virtual points can be expressed as linear functions of the pixel coordinates:

$$P_{\text{virtual}}^f = H_1 \begin{bmatrix} Z_s K_1^{-1} \mathbf{p}_1 \\ 1 \end{bmatrix} \quad (14)$$

$$P_{\text{virtual}}^b = T_{C2 \rightarrow C1} \cdot H_2 \begin{bmatrix} Z_s K_2^{-1} \mathbf{p}_2 \\ 1 \end{bmatrix} \quad (15)$$

Since $Z_s K_i^{-1}$ and the rigid-body matrices are all constant for a given configuration, each coordinate of $P_{\text{virtual}}^f = (x_f, y_f, z_f)^T$ and $P_{\text{virtual}}^b = (x_b, y_b, z_b)^T$ is an affine function of the pixel coordinates:

$$\begin{aligned} x_f &= A_1 u_1 + B_1 v_1 + C_1 \\ y_f &= D_1 u_1 + E_1 v_1 + F_1 \\ z_f &= G_1 u_1 + H_1 v_1 + I_1 \end{aligned} \quad (16)$$

$$\begin{aligned} x_b &= A_2 u_2 + B_2 v_2 + C_2 \\ y_b &= D_2 u_2 + E_2 v_2 + F_2 \\ z_b &= G_2 u_2 + H_2 v_2 + I_2 \end{aligned} \quad (17)$$

where the coefficients $A_1, B_1, \dots, I_2$ are determined by the composite matrix products.

The target point is

$$P_T = \begin{bmatrix} x_0 \\ y_0 \\ z_0 + l \end{bmatrix} \quad (18)$$

where $(x_0, y_0, z_0)^T$ is the tool-bar mounting position and $l = l_T$ is the tool-bar length.

B. Expanded Distance Expression

Substituting the affine expressions into (11)–(13), the measurement H becomes an explicit function of five variables:

$$H(u_1, v_1, u_2, v_2, l) = \frac{\|\vec{d} \times \vec{t}\|}{\|\vec{d}\|} \quad (19)$$

with

$$\vec{d} = \begin{bmatrix} (A_2 u_2 - A_1 u_1) + (B_2 v_2 - B_1 v_1) + (C_2 - C_1) \\ (D_2 u_2 - D_1 u_1) + (E_2 v_2 - E_1 v_1) + (F_2 - F_1) \\ (G_2 u_2 - G_1 u_1) + (H_2 v_2 - H_1 v_1) + (I_2 - I_1) \end{bmatrix} \quad (20)$$

$$\vec{t} = \begin{bmatrix} -A_1 u_1 - B_1 v_1 + (x_0 - C_1) \\ -D_1 u_1 - E_1 v_1 + (y_0 - F_1) \\ -G_1 u_1 - H_1 v_1 + l + (z_0 - I_1) \end{bmatrix} \quad (21)$$

Key observation. When l is fixed and only the pixel coordinates u_1, v_1, u_2, v_2 vary (e.g. due to pixel extraction noise), the resulting change ΔH is essentially independent of l . The absolute value of H depends on l , but the measurement uncertainty σ_H depends only on the pixel noise magnitude and the system coefficients $A_1, \dots, I_2$.

C. Error Propagation

By first-order Taylor expansion, the variance of H is

$$\sigma_H^2 \approx \sum_{q \in \mathcal{Q}} \left(\frac{\partial H}{\partial q} \right)^2 \sigma_q^2 \quad (22)$$

where $\mathcal{Q} = \{u_1, v_1, u_2, v_2\}$ (or equivalently $\{y_f, z_f, y_b, z_b\}$ after coordinate transformation), and σ_q is the standard deviation of each input parameter.

The partial derivatives can be computed analytically from (19) or numerically via finite differences.

D. Monte Carlo Validation

To validate the linear error propagation, we perform a Monte Carlo simulation: $N = 10,000$ samples are drawn with each pixel coordinate perturbed uniformly within $\pm 1 \mu\text{m}$ (equivalent pixel noise). The empirical standard deviation $\hat{\sigma}_H$ is compared with the analytical σ_H from (22).

E. Sensitivity Analysis

Each input parameter is perturbed by $\pm 1 \mu\text{m}$ independently, and the resulting $|\Delta H|$ is recorded. The variance contribution of each parameter is

$$\eta_q = \frac{\left(\frac{\partial H}{\partial q} \right)^2 \sigma_q^2}{\sigma_H^2} \times 100\% \quad (23)$$

V. Camera Calibration

Accurate camera calibration is a prerequisite for the DCPAM pipeline. This section describes the calibration of intrinsic parameters, extrinsic parameters, and the screen-plane distance Z_s .

A. Intrinsic Parameter Calibration

Each camera is modelled with the SIMPLE_RADIAL model: focal length f , principal point (c_x, c_y) , and a single radial distortion coefficient k . The intrinsic matrix is

$$K = \begin{bmatrix} f & 0 & c_x \\ 0 & f & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (24)$$

Calibration is performed using COLMAP's feature extraction and Structure-from-Motion pipeline on a set of checkerboard images captured from multiple viewpoints.

B. Extrinsic Parameter Calibration

The extrinsic parameters describe the pose of each camera relative to a common world frame W :

$$E_i = \begin{bmatrix} R_i & t_i \\ \mathbf{0}^T & 1 \end{bmatrix}, \quad i = 1, 2 \quad (25)$$

The relative transformation $T_{C2 \rightarrow C1}$ is then obtained via (8).

The dual-camera calibration procedure follows these steps:

- (1) Capture synchronised image pairs from both cameras.
- (2) Extract and match features across all images simultaneously.
- (3) Run SfM mapping with undistortion.
- (4) Export intrinsic and extrinsic parameters.
- (5) Compute relative pose (expected baseline ~ 7.5 – 8.0 cm).

C. Screen-Plane Distance Calibration

The distance Z_s from the camera optical centre to the receiving screen along the optical axis is a critical parameter, as it scales the back-projection (3). Z_s is determined by the mechanical dimensions of the probe fixture and verified by measurement.

The composite matrix $Z_s K_i^{-1}$ can also be calibrated directly by observing known points on the receiving screen.

D. Camera Tilt Angle Calibration

When the camera is mounted in a nominally horizontal orientation, a residual tilt angle α may exist between the camera's x -pixel direction and the horizontal direction of the world frame. This angle must be calibrated and compensated.

The relationship between a world-frame point (X, Y) and the camera-frame point (x, y) under a planar rotation is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} \quad (26)$$

VI. Mirror Transformation Calibration

The matrices H_1 and H_2 (Section III-E) encode the mirror rotation. Their calibration requires knowledge of: the rotation centre position R_{O1} (or R_{O2}); the rotation angle θ .

VII. DCPAM-CNN: Deep-Learning-Based Approach

While DCPAM-CV relies on hand-crafted feature extraction and manual parameter tuning, DCPAM-CNN trains a convolutional neural network to directly regress the laser spot centre coordinates from raw camera images.

TABLE I
Camera Calibration Parameters

Parameter	C_1 (Left)	C_2 (Right)
Model	SIMPLE_RADIAL	SIMPLE_RADIAL
Resolution (px)	2592×1944	2592×1944
Focal length (px)	2918.30	3757.72
Principal point	(1296, 972)	(1296, 972)
Distortion k	-0.2137	-0.2384

VIII. Tilt Deviation and Measurement Accuracy

In practice, the measurement probe may not be perfectly aligned with the nominal direction. A tilt deviation β introduces systematic error into the distance measurement. This section derives the relationship between β and the measurement accuracy, and determines the maximum allowable tilt for a specified accuracy target σ .

Given a target accuracy σ , the tilt rejection threshold $\beta(\sigma)$ is defined as the maximum inclination angle beyond which the measured data should be discarded to maintain the required precision.

IX. Experiments

A. Experimental Setup

The experimental platform consists of a Daheng Galaxy industrial camera pair (resolution 2592×1944), a collimated laser source, receiving screens with mirror mounts, and a precision linear stage for ground-truth positioning.

B. Calibration Results

Table I summarises the calibration results for both cameras.

C. Measurement Accuracy

X. Conclusion

This paper presented DCPAM, a dual-camera-based point-to-axis measurement model that replaces the traditional steel wire with a laser reference line. The complete mathematical pipeline—from pixel extraction through coordinate transformation to point-to-line distance computation—was derived and validated through error propagation analysis and Monte Carlo simulation.

Acknowledgments

References

- [1] J. Piotrowski, *Shaft Alignment Handbook*, 3rd ed. Boca Raton, FL, USA: CRC Press, 2006.